

Bharat Agrawal

✉ bharat.agrawal1322@gmail.com · 📞 +91 9368914189

[LinkedIn](#) · [GitHub](#) · [LeetCode](#) · [Codeforces](#) · [CodeChef](#)

EDUCATION

Indian Institute of Technology (IIT), Mandi

B.Tech. in Computer Science and Engineering (CGPA: 9.04 / 10.00)

Relevant Coursework: DSA, Database Systems, Machine Learning, Deep Learning, Computer Organization

Aug 2023 – Present

Mandi, India

Technical University of Munich (TUM)

Exchange Semester in Informatics (German Grade: 1.7/4.0, Equivalent GPA: 9.44/10.00)

Relevant Coursework: Graph Neural Networks, Advanced C++, Natural Language Processing

Oct 2025 – Mar 2026

Munich, Germany

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Bash, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Flask, Django, Streamlit, Pandas, NumPy, React.js, Node.js, FastAPI, Selenium, BeautifulSoup

Tools & Skills: AWS, OOP, Prompt Engineering, Agile, VSCode, Linux, Docker, Git, GitHub, FAISS, Perffetto

WORK EXPERIENCE

Google, Software Engineering Intern

May 2026 – Aug 2026

Silicon Product Software

Bengaluru, India

- Streamlined performance analysis for Linux and Android by architecting a unified profiling pipeline that synchronizes 490+ GPU hardware counters and CPU traces for GUI visualization, reducing bottleneck identification time by 40%.
- Identified 5+ GPU performance bottlenecks by developing custom scripts to merge kernel-space events with a newly implemented user-space API logging mechanism.
- Optimized GPU driver by tweaking heap memory allocation, reducing OOM stall cycles by 60%.
- Enhanced the GPU pipeline by replacing synchronous cache flushes with deferred background cache invalidations.
- Patched the ANGLE trace test to support headless Vulkan trace replays on Debian based platforms.

RESEARCH AND PROJECTS

Tabularis - Table Understanding and Query System

[GitHub](#)

Python, TensorFlow, HTML, Tailwind CSS, JavaScript, React.js, Next.js, Flask

Dr. Aditya Nigam, SCEE

- Developed a robust data extraction pipeline using a Preprocessor and Pix2Struct (240M), transforming complex PDF and image tables into structured HTML and CSV files.
- Processed 1000+ documents, achieving 95%+ accuracy in table recognition and data extraction.
- Integrated Qwen 3.0 with memory, enabling table queries across 50+ diverse types achieving 90% resolution rate.

SemEval-2026 Task-12: Abductive Event Reasoning for Large Language Models

[Paper](#)

Python, TensorFlow, Llama 3.2, Unsloth, BM25, Faiss, Google Colab

TUM, Munich

- Built a dual filtering pipeline (BM25 + FAISS re-ranking) to extract causal evidence from large document collections.
- Fine-tuned Llama 3.2 3B and 8B using 4-bit quantization (Unsloth, ~0.75% params trained).
- Llama 3.2 8B achieved accuracy 0.80, precision 0.90, F1 0.88, well above all few-shot prompting baselines.

Automatic Modulation Recognition

[GitHub](#)

Python, PyTorch, Streamlit, NumPy, Ngrok, Scikit-learn, Flask, MATLAB, VHDL

Dr. Robin Khosla, SCEE

- Implemented the VTCNN2 model using the Radio ML 2016.10a Dataset, achieving an overall accuracy of 72% in classifying 8 Digital and 3 Analog signal types of SNR 0-18 db.
- Proposed a DNN consisting of CNN, LSTM, GRU layers, enhancing classification accuracy to 92%.
- Reduced model complexity by downsizing to 106k params, decreased inference latency by 40% with an average response time of 50 ms.

ACHIEVEMENTS

- Secured 2nd place in HCLTech Hackathon organized under CS671 course.
- Ranked 10th nationwide among 50+ teams in CSAW CTF.
- Attained 4th rank in Scythe CTF and 6th rank in Chronos CTF.
- Solved 1100+ coding problems across platforms like Codeforces, LeetCode, CodeChef, GFG.

POSITIONS OF RESPONSIBILITY

- Mentor, Kamand Prompt - Programming Club of IIT Mandi
- Mentor, Association for Computing Machinery (ACM)